

Rack Management System

Software Requirements Specification (SRS)

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Module	Warehouse / Inventory – Rack Management
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1. Introduction

1.1 Purpose

This document defines the functional and data requirements for the Rack Management System — a module used to track physical storage racks in a warehouse and to record which items, colors, sizes and quantities are stored in each rack. It converts the source notes into a structured specification that developers and testers can use to build and validate the module.

1.2 Scope

The system shall allow users to: maintain a master list of racks with their capacity and occupancy status; assign inventory items (with color, size and quantity) to a specific rack location; update or remove stock from a rack; move stock between racks; and generate a rack-wise report showing what is stored in each rack.

1.3 Definitions & Naming Convention

Each rack location is identified using a hierarchical Rack ID in the format **R{Rack}-S{Shelf}-B{Bin}**, for example **R05-S02-B04** refers to Bin 04, Shelf 02, of Rack 05. This allows a single rack structure to be broken down into shelves and bins for granular location tracking.

2. Data Model

2.1 Rack Master Table

Stores the master record of every physical rack location, its total capacity, quantity currently used, and its occupancy status.

Field	Data Type	Description
Rack ID	String (PK)	Unique identifier for the rack, in Rack-Shelf-Bin format (e.g., R05-S02-B04).
Capacity	Integer	Maximum quantity the rack location can hold.
Used	Integer	Quantity currently occupied in the rack.
Status	Enum	Derived/assigned status of the rack: Vacant or Occupied.

Sample data:

Rack ID	Capacity	Used	Status
R001	100	80	Occupied
R002	100	0	Vacant
R003	100	45	Occupied
R004	100	0	Vacant

2.2 Item Location Table

Stores each item's placement details and links back to the Rack Master table via a foreign key.

Field	Data Type	Description
Item	String	Name/code of the item.
Color	String	Color variant of the item.
Size	String	Size variant of the item.
QTY	Integer	Quantity of this item/color/size placed in the rack.
fkRackId	String (FK)	Foreign key referencing Rack Master.Rack ID.
moduleId	String/Integer	ID of the source transaction that generated this record.
moduleType	Enum	Source module of the transaction. Allowed values: Purchase Inward, Job Slip, Pack Design, Sales Return.

3. Functional Requirements

3.1 Add Item to Rack

The system shall support two alternative flows for assigning an item to a rack:

Flow A — From a source transaction/module:

1. User selects the source Module (Purchase Inward / Job Slip / Pack Design / Sales Return).
2. Item details (Item, Color, Size, QTY) are auto-filled from the selected transaction.
3. User selects a Rack from the Rack Master list, either by searching or by manual entry.
4. User saves the record; the system creates an Item Location entry and updates the rack's Used quantity and Status.

Flow B — Manual entry:

1. User adds the Item.
2. User adds the Color.
3. User adds the Size.
4. User adds the QTY.
5. User selects a Rack from the Rack Master list, either by searching or by manual entry.
6. User saves the record; the system creates an Item Location entry and updates the rack's Used quantity and Status.

3.2 Update / Remove Rack Item

The system shall allow the user to update the quantity of an existing item stored in a rack (including reducing it to zero to remove it from the rack) and save the change. On save, the system shall recalculate the associated rack's Used quantity and Status.

1. User locates the item's existing rack record.
2. User updates the QTY value.
3. User saves; the system updates the Item Location record and recalculates Rack Master Used/Status.

3.3 Move Rack Item

The system shall allow items to be moved from one rack to another.

1. User opens the Rack Management list.
2. User selects the source Rack.
3. User specifies the QTY to move.
4. User saves; the system deducts the moved QTY from the source rack and adds it to the destination rack, updating Used and Status for both rack records.

3.4 Rack-Wise Report

The system shall provide a report view that can be searched/filtered by Rack, showing all items currently stored in the selected rack (Item, Color, Size, QTY, source module).

4. Business Rules & Validations

- A rack's **Status** shall be set to **Vacant** when Used = 0, and **Occupied** when Used > 0.
- A rack's **Used** quantity shall never exceed its **Capacity**; the system shall block save actions that would breach capacity.
- **moduleType** on the Item Location table shall be restricted to one of: Purchase Inward, Job Slip, Pack Design, or Sales Return.
- Rack IDs shall follow the R{Rack}-S{Shelf}-B{Bin} naming convention to uniquely identify a bin-level storage location.
- Moving an item between racks shall be treated as an atomic operation — both the source deduction and destination addition must succeed together, or the entire move must be rolled back.
- Rack selection in all flows shall support both searching the Rack Master list and manual entry of a Rack ID.

5. Assumptions & Open Points

The following points were not fully specified in the source notes and are flagged for confirmation with stakeholders before development:

- Whether Capacity/Used/QTY are tracked as unit counts or another measure (e.g., weight, cartons).
- Whether partial moves/updates require additional approval or audit logging.
- Whether a rack can hold multiple different items simultaneously, and if so, how Capacity is shared across them.
- User roles/permissions for who can Add, Update, or Move rack items.

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